RAI Risk Taxonomy 2.0: Top-down evidence selection and staged classification of 1,711 global AI risks

RAI Risk Taxonomy Project

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Abstract

We present a reproducible, top-down account of the evidence and classification pipeline used to construct RAI Risk Taxonomy 2.0. The integrated evidence space contains 3,906,767 AI records: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. A curated AI-risk evidence view contains 82,971 records (2.1% of the AI evidence space). Canonicalization, source crosswalking, and risk-card construction yields 1,726 candidate rows. A conservative exact-match check merges one duplicate, yielding 1,725 candidate L4 cards. A taxonomy-level check against the 24 authoritative Physical AI L3 labels excludes 14 exact label leaks, yielding 1,711 L4 cards. A staged hierarchy-blind classification procedure assigns all 1,711 cards (100.0%) to 50 L3 nodes; 76 cards (4.4%) remain marked `HOLD`. We specify the filter at each transition, the mathematical decision rules, the protected Physical AI set, and the limits of interpreting record-level evidence as card-level taxonomy coverage.

Keywords: AI risk taxonomy; evidence synthesis; open-set classification; semantic mapping; Physical AI; responsible AI.

1 Introduction

A taxonomy can appear complete while concealing uncertainty in its source corpus, unit of analysis, or placement rules. RAI Risk Taxonomy 2.0 therefore separates four quantities that must not be conflated: the size of the integrated AI evidence space, the size of the AI-risk evidence view, the number of canonical L4 risk cards, and the number of L4 cards assigned to L3. The central result is a top-down evidence-to-taxonomy chain of 3,906,767 records, 82,971 AI-risk records, 1,726 candidate L4 rows, 1,725 exact-deduplicated candidates, 1,711 level-validated L4 cards, and 1,711 L3-assigned cards.

The transitions do not all preserve the same statistical unit. The first two stages count documents or patent records. The third and fourth stages count canonical risk concepts. Consequently, the 82,971 evidence records are not a one-to-one parent table from which every L4 card was mechanically generated. They form an auditable evidence view supporting corpus coverage and risk interpretation, whereas the L4 registry integrates risk taxonomies, benchmarks, standards, and linked evidence through a source crosswalk.

2 Top-down dataset construction

2.1 Stage A: integrated AI evidence space

The top-level dataset contains 3,906,767 records from three source families: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. Paper records originate from the local global Web of Science AI collections. Policy records combine the reviewed curated AI-policy master with the broader policy-materials collection. Patent records originate from the stage-2 AI patent master.

Records are standardized to a common schema and retained only when a non-empty title is available. Deduplication is performed within document type and analytical category. The ordered key hierarchy is patent application ID for patents, followed by DOI, source record ID, URL, and a normalized title–year–institution fallback. Cross-category overlap is intentionally retained because AI, AI infrastructure, Physical AI, and AI risk are analytical views rather than mutually exclusive partitions.

2.2 Stage B: integrated AI-risk evidence view

The AI-risk evidence view contains 82,971 records: 50,375 papers, 938 policy reports, and 31,658 patents. Its inclusion rule is source-governed rather than a single keyword query. Papers combine curated academic risk references, the clean and integrated AI-risk paper corpora, Physical AI safety benchmarks, and verified paper records from the top-200 evidence set. Policy reports combine curated risk-policy references with verified report records. Patents combine curated risk-patent references with verified patent records. Record-type filters are applied before the same title-validity and within-category deduplication rules used at Stage A.

This view represents 2.1% of the integrated AI evidence space. Because the source collections and analytical categories can overlap, this percentage is a descriptive ratio, not the estimated prevalence of risk research in all AI scholarship.

2.3 Stage C: canonical L4 registry

The unit changes from evidence record to risk concept. Source risk IDs are joined through the source crosswalk, semantically redundant cards are reviewed, and one canonical row is retained per L4 failure mode or harm mechanism. Permanent identifiers `RAI4-0001` through `RAI4-1726` are allocated independently of the risk label. The initial result is 1,726 candidate L4 rows. Automatic deduplication is restricted to pairs whose normalized label and mechanism-only definition are both exactly equal; semantic-similarity thresholds are not used. This merges `RAI4-1606` into `RAI4-1083`, retains both source references, and yields 1,725 exact-deduplicated candidates. The retired ID remains in the crosswalk. The registry includes the 182-card Physical AI taxonomy as a protected mapping set.

The ratio of L4 cards to AI-risk evidence records is 2.1%. It is reported only as a scale-change diagnostic: one document may support multiple risks, and one risk may be supported by multiple documents. It is not a document acceptance rate.

2.4 Stage D: taxonomy-level validation

The deployed Physical AI site is the authoritative source for its 24 L3 names and 182 L4 cards. Its live HTML was byte-identical to the local repository snapshot used for this check. We compare the English component of each authoritative Physical L3 name with every candidate L4 label after NFKC normalization, case folding, ampersand normalization, and removal of non-alphanumeric separators. A card is excluded only when these normalized labels are exactly equal; semantic similarity and definition similarity are not used. Fourteen L4 rows match category names and are retired as taxonomy-level leakage, including `RAI4-0553` “Accidental harm”, which maps to Physical L3 `P3.1`. Their definitions and references remain in the audit crosswalk. This yields 1,711 canonical L4 cards without changing any of the protected 182 Physical L4 mappings.

2.5 Authoritative Physical L4 content synchronization

Release v2.4 synchronizes the content of all 182 protected Physical cards from the local repository underlying the deployed Physical AI Risk Taxonomy site. The immutable RAI4 crosswalk and L3 paths are retained, while the bilingual label and definition, severity, probability, 3H1R attributes, evidence justifications, and 360 reference rows are replaced by the current Physical source values. Of these reference rows, 359 contain a justification and one source row retains an empty justification without imputation. Bilingual terminal parenthetical text is split into Korean and English fields to preserve the global card schema. Impact is recalculated as $I = s\,p$ from the synchronized severity s and probability p. Percentile is not displayed for any card because the cards do not constitute a common measurement population suitable for relative ranking. Existing percentile fields remain provenance data only.

2.6 Complete English–Korean localization

Release v2.5 requires bilingual display at every taxonomy level. The supplied L0–L3 terminology and Korean definitions are used as the controlled glossary. The 182 Physical L4 cards retain their authoritative bilingual source text. For the remaining 1,529 cards, the localization pipeline removes the repeated provenance boilerplate from each English definition, selects the first sentence that states the core failure mechanism, generates a concise Korean definition, and normalizes risk-domain terminology. Context-sensitive corrections include *poisoning* as "data or model contamination" rather than substance addiction, *washing* as false compliance presentation, and *rubber stamping* as nominal approval. Every generated row retains the English source-sentence hash, method, and `human_review_status=pending`. Thus all 1,711 cards contain non-empty English and Korean labels and definitions, while automated Korean drafts remain explicitly reviewable rather than being represented as human-approved ground truth.

2.7 Canonical L2 consolidation

Release v2.6 normalizes L2 to three canonical categories: *Interaction Safety*, *System Safety*, and *Societal Safety*. The former General labels *Interaction* and *System*, and the former Agentic label *System*, are renamed accordingly. Six L1-specific path nodes remain as implementation edges so that the strict L0–L1–L2–L3 tree and all 50 existing L3 parent paths are preserved; each edge references one of the three canonical L2 identifiers. Thus the reported L2 category count is three, while no L3 path or L4 assignment changes.

2.8 Stable L4 identifiers and L1 display semantics

The public explorer keeps the stable `RAI4-####` identifier independent of taxonomy placement. L1 membership is communicated through an explicit text badge, a consistent icon, and a visual color: a brain and blue for General-purpose AI, a compass and green for Agentic AI, and a robot and red for Physical AI. The icon–color pair is repeated in the global navigation, filters, hierarchy panel, and L4 cards. The color is derived from the card's current L1 path at render time rather than encoded into the permanent identifier. This allows later human-approved remapping without identifier churn and avoids using color or icon as the sole carrier of categorical information.

2.9 Stage E: L3 assignment policy

The final policy view assigns all 1,711 canonical L4 cards to one of 50 L3 nodes (100.0%). After taxonomy-level validation, 76 cards (4.4%) retain a `decision_required` marker, displayed as `HOLD`. This marker is metadata on an assigned card, not an L1, L2, or L3 taxonomy node.

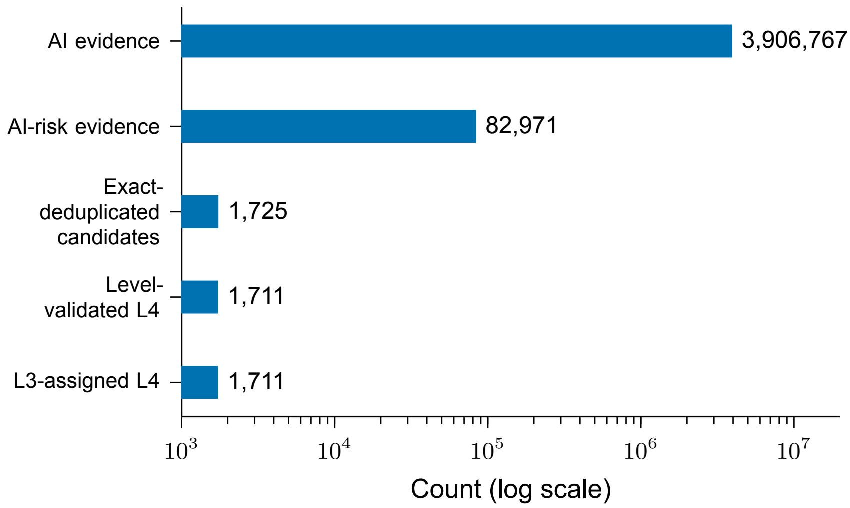

Figure 1: **Top-down evidence and taxonomy funnel.** Counts move from document/patent records to canonical risk concepts at the transition from AI-risk evidence to L4. The logarithmic axis preserves visibility across four orders of magnitude.

Table 1: Top-down stages, counts, and inclusion filters.

Stage	Count	Unit	Principal filter
AI evidence	3,906,767	Document or patent record	AI source collections; valid title; standardized schema; within-type/category deduplication.
AI-risk evidence	82,971	Document or patent record	Curated/integrated risk sources; record-type inclusion; valid title; within-type/category deduplication.
Exact-deduplicated candidates	1,725	Risk card	Exact normalized label and core-definition match only; one retired ID mapped to its canonical card.
Canonical L4	1,711	Risk card	Exact normalized-label exclusion against 24 authoritative Physical L3 names; 14 level leaks retired; all 182 Physical L4 cards preserved.
L3 assigned	1,711	Risk card	Protected Physical mapping or staged non-Physical placement; 76 decision-required cards remain assigned and marked `HOLD`.

3 Mathematical classification procedure

3.1 Protected Physical AI mapping

The 182 L4-to-L3 mappings inherited from the Physical AI taxonomy are treated as the locked gold set. Later stages cannot change these paths. Before the taxonomy-level exclusion, the remaining 1,543 candidates enter the hierarchy-blind classification procedure. Legacy global L0–L3 labels, source families, and source-ID prefixes remain reference metadata and are excluded from prediction features.

3.2 Stage 1: strict open-set gate

The input for each non-Physical card is the L4 label, a mechanism-only definition with hierarchy restatements removed, and the evidence title. Let s_{sem} be the top-1 semantic score, s_{comp} the top-1 composite score, m_{sem} and m_{comp} their respective top-1 margins, and v_{anchor} the number of anchor views selecting the same L3. A candidate is retained only when all seven conditions hold:

1. exactly one L3 is rule-eligible;
2. the rule winner, semantic top-1, and composite top-1 are identical;
3. $s_{sem} \geq 0.55$ and $s_{comp} \geq 0.54$;
4. $m_{sem} \geq 0.035$ and $m_{comp} \geq 0.035$;
5. $v_{anchor} \geq 2$ across three anchor views;
6. no gap sentinel, multi-mechanism flag, or Physical-boundary flag is present; and
7. two independent expert agents approve the same exact L3.

This gate yields 37 new expert-consensus proposals. Together with the 182 Physical locks, Stage 1 assigns 219 cards and withholds 1,507.

3.3 Stage 2: suitability-ranked relaxation

Stage 2 preserves all 219 protected assignments and computes

$$S_2 = 0.55q_{sem} + 0.20q_{sm} + 0.10q_{cm} + 0.10q_{vote} + 0.05q_{rule} - p_{gap}, \tag{1}$$

where

$$q_{sem} = \text{clip}\left(\frac{s_{sem} - 0.45}{0.25}, 0, 1\right), \tag{2}$$

$$q_{sm} = \text{clip}\left(\frac{m_{sem}}{0.05}, 0, 1\right), \tag{3}$$

$$q_{cm} = \text{clip}\left(\frac{m_{comp}}{0.05}, 0, 1\right), \tag{4}$$

$$q_{vote} = v_{anchor}/3, \tag{5}$$

$$q_{rule} = \mathbf{1}[\text{at least one eligible L3}], \tag{6}$$

$$p_{gap} = 0.05\,\mathbf{1}[\text{gap sentinel present}]. \tag{7}$$

S_2 is a deterministic ranking score rather than a calibrated probability. Stage 2 first preserves 55 hard holds. It then orders the remaining holds by ascending S_2, with permanent RAI4 ID as the tie breaker, and reserves the lowest 118. Thus $55+118 = 173$ cards remain unresolved and 1,553 cards are assigned (90.0%).

3.4 Stage 3: forced-path test and final policy

Stage 3 preserves the 1,553 Stage 2 paths and assigns only the 173 unresolved cards to the hierarchy-blind top-1 L3 already stored in the score artifact. This produces 1,726 operational paths but does not validate taxonomy fit. Each new path retains its original hold reason, score, review priority, and `human_approved=false` marker.

The final policy marks five difficult-case families for decision review: 23 Physical-scope reviews, 17 expert rejections, 7 expert disagreements, 6 multi-mechanism cases, and 2 low-absolute-fit cases. Their forced L3 path remains populated and `decision_required=true`. The remaining 118 quota holds also retain their forced paths and review metadata.

3.5 Independent expert card review

Two reviewers independently inspected all 1,725 cards for agreement among the L4 label, mechanism definition, evidence, and assigned L3. Reviewer A proposed 30 amendments and Reviewer B proposed 14. Automatic amendment required both reviewers to propose the same label and L3 at high confidence. One card met this rule: `RAI4-0888` changed from “Privacy concerns” to “Anthropomorphic trust-induced privacy disclosure”, while retaining `RAI3-G-INT-10`. Its original label remains in provenance metadata. Forty-two cards had a proposal from only one reviewer or incompatible proposals; their labels and paths were not changed. Thirteen were already marked `HOLD` and 29 newly received the marker. The consensus amendment cleared the prior hold on `RAI4-0888`, giving $55 - 1 + 29 = 83$ final `HOLD` cards. No protected Physical AI card was changed.

The later taxonomy-level validation supersedes the proposed adjudication of `RAI4-0553`: “Accidental harm” is itself the authoritative Physical L3 category `P3.1`, not an L4 failure mode. It is therefore excluded rather than renamed or remapped. Thirteen additional exact Physical-L3 label leaks are handled identically. Six of the excluded rows previously carried `HOLD`, reducing the displayed count from 82 to 76. The protected Physical set remains exactly 182 cards.

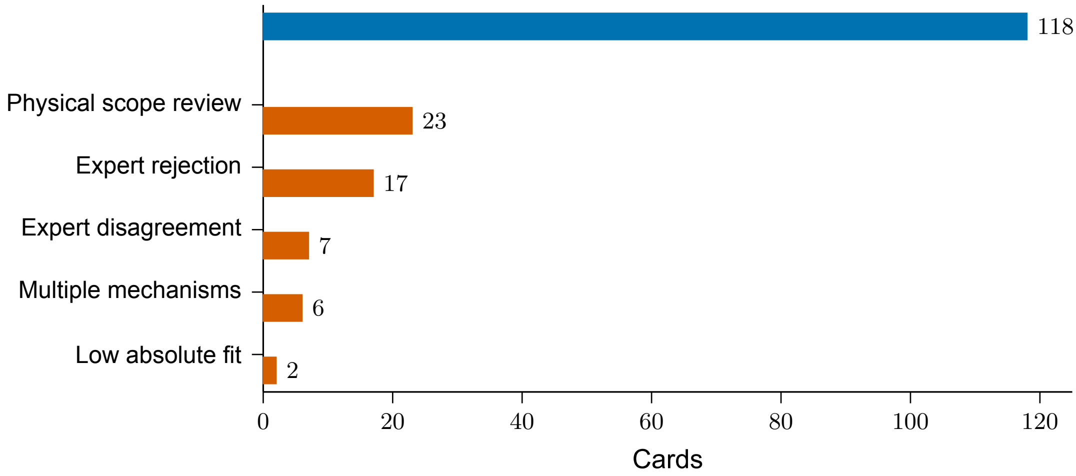

Figure 2: **Disposition of the 173 Stage 2 holds.** The 55 difficult cases (vermillion) are force-assigned and marked `HOLD`; 118 suitability-reserve cards (blue) are also force-assigned.

4 Results

4.1 Evidence composition

Papers account for 44.8% of the integrated AI evidence space, policy reports for 3.2%, and patents for 52.0%. Within the AI-risk evidence view, the corresponding shares are 60.7%, 1.1%, and 38.2%. These shares use one decimal because their integer component is at least one.

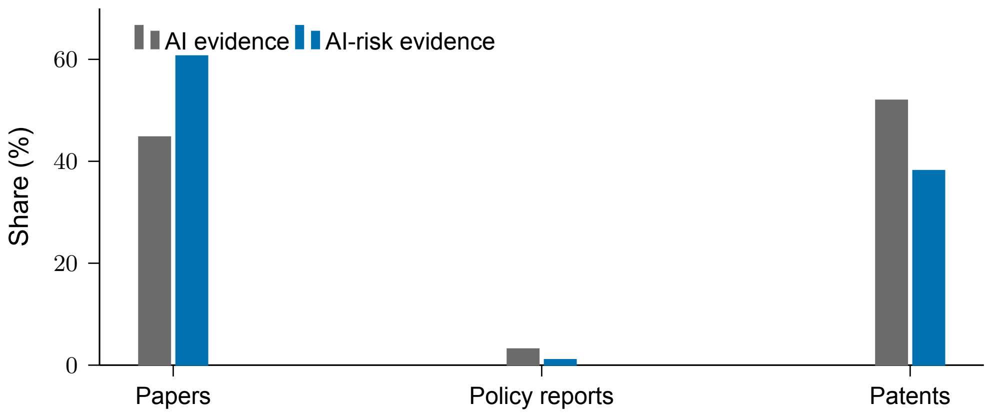

Figure 3: **Source-family composition.** Percentages compare document-type composition within the integrated AI evidence space and the AI-risk evidence view.

4.2 Final hierarchy coverage

The final policy assigns all 1,711 canonical cards (100.0%); 76 assigned cards (4.4%) carry the `HOLD` marker. Of all cards, 1,180 are General, 349 are Agentic AI, and 182 are Physical AI. Their shares are 69.0%, 20.4%, and 10.6%, respectively. All 50 L3 nodes contain at least one card. Anthropomorphism is the largest L3 with 247 cards (14.4%); the five largest L3 nodes jointly contain 41.9%.

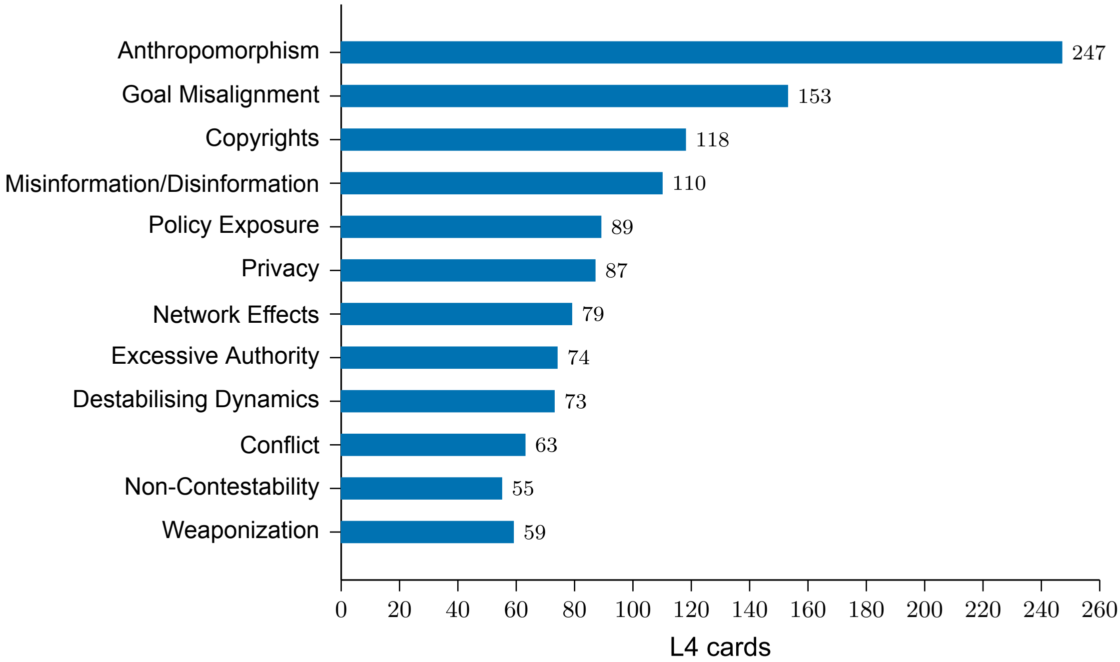

Figure 4: **Largest L3 nodes in the final policy view.** The figure shows the twelve largest L3 nodes. Counts include the 76 assigned cards marked `HOLD`.

5 Rounding and reporting policy

All counts are reported as integers. Percentages below 1 are rounded at the third decimal place and displayed to two decimal places; percentages with an integer component of at least 1 are rounded and displayed to one decimal place. For example, the L4 registry represents 0.04% of the integrated AI evidence count, the AI-risk view represents 2.1%, final L3 coverage is 100.0%, and the decision-required share is 4.4%. Exact floating-point ratios remain available in the machine-readable statistics artifact.

6 Validation and limitations

The 182 Physical mappings and 37 Stage 1 expert-consensus mappings remain unchanged in later stages. Repeated Stage 2 and Stage 3 builds produce byte-identical placement hashes. The Stage 2 validator reports 13 PASS, 0 FAIL, and 3 WARN; the Stage 3 validator reports 14 PASS, 0 FAIL, and 4 WARN. The release validator reports 71 PASS, 0 FAIL, and 2 WARN, and all 32 repository tests pass at the time of this report revision.

Three limitations directly affect interpretation. First, cross-category overlap is retained in the integrated evidence space, so category counts are analytical views rather than mutually exclusive samples. Second, the transition from 82,971 evidence records to 1,711 canonical L4 cards changes the unit of analysis and is not a mechanical screening step. Third, 1,214 of the 1,334 new Stage 2 placements lack Stage 1 rule eligibility, 247 override a taxonomy-gap sentinel, and the remaining 118 Stage 3 forced candidates are not human-approved ground truth.

7 Review priorities

The 76 assigned cards marked `HOLD` should be reviewed first, beginning with the 42 cards lacking two-reviewer consensus and the original hard-hold cases. The 118 quota candidates should then be reviewed in ascending S_2 order. Finally, the 247 gap overrides and 1,214 rule-free placements should be audited through L3-stratified samples. Human review should determine whether each failure is best addressed by expanding an L3 definition, introducing a new L3, or allowing primary and secondary labels.

Data availability

The exact top-down counts, filters, placements, and public card bundle are available in the repository:

- `reports/statistics/top_down_evidence_funnel.json`
- `data/experiments/stage3-forced-policy-v2/placements.json`
- `public/data/releases/v2.6.0/cards.json`
- `reports/data_quality/l4_deduplication_v2.1/retired_to_canonical.json`
- `reports/expert_review/v2.1/consensus_result.json`
- `reports/data_quality/physical_l3_label_leakage_v2.3.json`
- `reports/data_quality/physical_card_sync_v2.4.json`
- `reports/localization/nonphysical_l4_ko_v2.5.json`
- `reports/data_quality/l2_category_consolidation_v2.6.json`

This document is the English journal-style edition of *RAI Risk Taxonomy Technical Report 2.0*.